

EDUCATION AND TRAINING:

Bachelor of Engineering- Software Engineering – GPA (4.07/4.3)

August 2020- April 2025

Ontario Tech University – Oshawa, ON.

Courses: Web Programming, Java OOP and Design, Software Design, Data Structure & Algorithms, Data Management

SKILLS:

Languages: Java, JavaScript, Python, C#, C++, PHP, SQL, HTML, CSS

Technologies: Git, ReactJS, Redux, Bootstrap, MySQL, MongoDB, Node.js, Express.js

WORK AND VOLUNTEER EXPERIENCE:

Undergraduate Research Assistant

May 2022 – September 2022

Ontario Tech University- [SAIN Lab](#), Oshawa, ON

- Worked with a highly qualified team of researchers and professors to increase awareness of Password Managers and the effect of having a weak password. Presented relative papers, ideas, and solutions to the team.
- Developed a full stack survey website using **JavaScript, HTML, CSS, ExpressJS, Node.js, MongoDB** to learn about and spread awareness on the use and risks of Password Managers.
- Wrote various **python scripts** to analyze results from pilot testers and wrote **documentation** on the entire process.
- The survey will be taken by over **1000 people** on Amazon Mechanical Turk and will effectively increase awareness of password managers and risks of weak passwords by over **50%**.

Software Engineering Peer Tutor

August 2021 – May 2022

Ontario Tech University, Oshawa, ON

- Conducted Physics study hall and one on one tutoring sessions to tutees in **Introduction to programming (C++)**, **Data Structures & Algorithms (Python)**, **Web Programming** and **Object-Oriented Programming (Java)**.
- Guided students down the path of independent learning and encouraged active participation leading to tutees understanding programming concepts well.
- Learned to better explain technical concepts, give detailed feedback, and received **Positive feedback** from students.

Undergraduate Research Assistant

May 2021 – August 2021

Ontario Tech University- [AD2MLabs](#), Oshawa, ON

- Developed a program using **Python** that reads through an ASCII STL file and stores triangle information into a multi layered data structure. This program will be used for further research by PhD students.
- Developed a CAD GUI using **PyQt** and **Qt Designer** that displays an STL file in 3D, with many user functionalities such as seeing triangle information, to be used in further research to repair STL files.
- Developed **three websites** ([ahmadbarari.com](#), [ad2mlabs.com](#), [robin-team.com](#)) using WordPress, **HTML and CSS**, increasing my understanding of front end developing and **UX design**.

PROJECTS:

Content Based Image Retrieval - [GitHub](#)

April - 2022

- Developed a **Python** program that converts a black and white image into a barcode using angled projections.
- The program can then search through a list of input Images or barcodes to find the most similar ones with an accuracy of **>80%**.
- Wrote a research report on the analysis of the program and **data structures** used. Received a grade of **100%**

Online Grocery Store – [GitHub](#) | [Website](#)

December 2021

React, JavaScript, CSS, ExpressJS

- Led team of 4: Organized meetings, delegated tasks, helped teammates debug, etc., leading to a final grade of **100%**
- Designed responsive online grocery store, implemented an authentication system, and built Home, Product & Cart Pages. Implemented dynamic cart functionality using **Redux**.